

NEO Mathematics Lesson Guidance

Ratio and Proportion - Scaling Recipes

Driving Question: Can maths help us share, scale and sustain things fairly?

Lesson at a Glance

Mathematical focus: Ratio is fundamentally about relationships between quantities. The key conceptual shift is from additive thinking to multiplicative thinking: if the number of people becomes three times as large, every ingredient becomes three times as large.

Stepping Stones: use ratio language and notation; recognise connected quantities; use scale factors; scale quantities multiplicatively; complete ratio tables; explain why equivalent ratios preserve a relationship.

Primary NEO Cornerstones: Connection, Movement and Nutrition.

Suggested duration: 45-75 minutes, or more than one shorter session. Timings are invitations, not deadlines. Follow the learner's regulation, engagement and mathematical readiness.

The lesson does not need to be completed in one sitting. A learner may speak, type, draw, use the Scratchpad, work on paper or pause and return.

Supporting the mathematics

Ratio is fundamentally about relationships between quantities. The key conceptual shift is from additive thinking to multiplicative thinking: if the number of people becomes three times as large, every ingredient becomes three times as large.

Resources and materials

Essential

- A device capable of opening the interactive HTML lesson
- Paper or a notes app
- Calculator where appropriate to the learner's pathway

Helpful

- Measuring jug
- Kitchen scales
- Cups or containers
- Counters
- Sticky notes

- Blank table paper

Optional extension or context

- Real or empty food packaging
- Recipe cards
- Price information supplied by the supporting adult
- Coloured pens. No physical food preparation is required. Consider allergies, dietary needs and sensory preferences if real food is used.

Suggested lesson journey

Arrive and Connect - about 5 minutes

Read the Driving Question. There is no required correct opening response. Ask what 'fair' might mean, whether the learner has ever changed a recipe, or what might happen if only one ingredient was doubled.

Explore the Recipe Scaler - 10-15 minutes

Allow the learner to change group size several times before teaching a procedure. Invite a Scratchpad prediction, then test it with the scaler. Ask what moves together and what stays unchanged.

Build a Ratio Table - about 10 minutes

Encourage the learner to find the multiplier before calculating every ingredient. Accept a valid less-efficient method before discussing efficiency.

Pause - flexible

The Rest section is part of the lesson. Look away from the screen, stretch, take a drink, write only the multiplier, or return later.

Create - 15-20 minutes

Use the Smoothie Design Challenge. Focus on base quantities, group size, scale factor, consistent scaling and mathematical explanation. Do not judge food choices.

Practise and Reflect - flexible

Use all eight practice questions or select a sample. The final portfolio piece may be completed now or in another session.

Common misconceptions and supportive responses

Additive reasoning

A learner may say that four extra people means four extra bananas.

Possible prompt: How many times bigger is 6 than 2? Can you show that multiplier on the Scratchpad?

Scaling only one quantity

The learner doubles the milk but leaves another ingredient unchanged.

Possible prompt: Would this still be the same recipe? Which quantities need the same multiplier?

Focusing on a constant difference

The learner notices the difference between 2 people and 1 banana rather than the multiplicative relationship.

Possible prompt: Compare 2:1, 4:2 and 6:3. Is the difference staying the same? What relationship is staying the same?

Cornerstone guidance

Connection

Mathematical role: Quantities remain related.

Notice or invite: Notice language such as 'when this doubles, that doubles'.

Movement

Mathematical role: Quantities scale up and down.

Notice or invite: Invite a prediction before changing the interactive.

Creativity

Mathematical role: The learner constructs a proportional recipe.

Notice or invite: Accept different valid ingredient choices and representations.

Reflection

Mathematical role: The learner distinguishes additive and multiplicative reasoning.

Notice or invite: Treat mistakes as objects to inspect, not evidence of failure.

Rest

Mathematical role: The problem can be separated into multiplier and quantities.

Notice or invite: Use the calm routine: start amount, new amount, multiplier, scale, check.

Nutrition

Mathematical role: Food and quantities ground proportional reasoning in sustaining a group.

Notice or invite: Avoid moral language such as 'good food' and 'bad food'.

Regulation, access and re-engagement

If the learner does not want to speak: allow typed responses, pointing, changing interactive values, drawing or private Scratchpad work. Watching and testing values can still be mathematical engagement.

If the learner is worried about being wrong: invite a prediction rather than an answer. Use language such as, 'Let's test that idea' or 'What did the graph/recipe show us?'

If the learner becomes overwhelmed: reduce the visual field, close other panels, use the Rest routine, work with one quantity or one graph feature, or pause the session.

If there are gaps in prior knowledge: use the interactive as a concrete starting point. Do not require fluency before exploration. Record the gap as useful curriculum information rather than failure.

If the learner finishes quickly: invite variation, comparison, proof, model limitations or a learner-generated example rather than simply more of the same.

Evidence and mastery

Emerging - Notices that quantities are connected and scales some values with support.

Developing - Uses a given scale factor and completes familiar proportional calculations.

Secure - Independently identifies scale factors and explains why all quantities use the same multiplier.

Mastering - Chooses efficient approaches, applies proportional reasoning in unfamiliar contexts and explains the invariant relationship.

Possible evidence includes an explanation, interactive choices, a Scratchpad image, completed table, verbal teach-back, corrected misconception, model or final portfolio piece.

The NEO Maths Scratchpad

The embedded Scratchpad is a thinking space. It is not a compulsory evidence task. The learner can use blank paper, square grid, coordinate grid or a lesson-specific template. Undo, redo and eraser controls are intentionally available so that revising thinking feels normal.

Suggested facilitation language

Notice first. Ask second. Explain only as much as the learner needs to keep thinking.

Useful prompts include: 'What are you noticing?', 'What stayed the same?', 'What changed?', 'Can you predict before we test?', 'What would convince you?', and 'Where might this model stop working?'